White's Conjecture and Stressed Subset Relaxations

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Roadmap

Symmetric exchange property

Matroid

A **matroid** is a set system $\mathsf{M} = (E, \mathcal{B})$ with

(B1) $\emptyset \neq \mathcal{B} \subseteq \binom{E}{r}$,

(B2) $\forall A, B \in \mathcal{B},\ \forall a \in A \setminus B,\ \exists b \in B \setminus A$ such that $A - a + b \in \mathcal{B}$.

The basis exchange axiom (B2) can be strengthened as follows:

Symmetric exchange property [Oxl06]

$$\forall A, B \in \mathcal{B},\ \forall a \in A \setminus B,\ \exists b \in B \setminus A$$

such that

$$\begin{aligned} A &\quad \overset{a \quad b}{\longleftrightarrow} \quad A - a + b \in \mathcal{B}, \\ B &\quad \overset{b \quad a}{\longleftrightarrow} \quad B - b + a \in \mathcal{B}. \end{aligned}$$

White's conjecture

Two **degree** m tuples of bases $\mathbf{B} = (B_1, \dots, B_m)$ and $\mathbf{B}' = (B'_1, \dots, B'_m)$ are **compatible** if $B_1 \uplus \cdots \uplus B_m = B'_1 \uplus \cdots \uplus B'_m$.

White's conjectures [Whi80]

Non-commutative (strong)

$$\mathbf{B}, \mathbf{B}' \text{ compatible} \iff \mathbf{B} \sim_{\mathrm{SE}} \mathbf{B}',$$

where $\sim_{\mathrm{SE}}$ is generated by *symmetric exchange* moves

$$(A, B) \mapsto (A - a + b,\ B - b + a).$$

Quadratic (weak)

$$\mathbf{B}, \mathbf{B}' \text{ compatible} \iff \mathbf{B} \sim_{\mathrm{Q}} \mathbf{B}',$$

where $\sim_{\mathrm{Q}}$ is generated by *quadratic* moves

$$(A, B) \mapsto (A', B'), \qquad A \uplus B = A' \uplus B'.$$

Positive results over time

Year	Matroid class	Reference
2008	Graphic	Blasiak (Combinatorica)
2010	Rank at most 3	Kashiwabara (E-JC)
2011	Lattice path	Schweig (J. Pure Appl. Algebra)
2013	**Sparse paving**	Bonin (Adv. Appl. Math.)
2014	Strongly base-orderable	Lasoń–Michałek (Adv. Math.)
2024	**Split (degree 2)**	Bérczi–Schwarcz (SIDMA)
2025	**Sparse paving revisited**	Han–Michałek–Weigert
2025	**Paving**	**Yu–Yuen**

Single-basis modification (Han–Michałek–Weigert)

For matroids M, M' with $\mathcal{B}(\mathrm{M}') = \mathcal{B}(M) \sqcup \{B'\}$, then

$$\mathrm{M} \text{ is White-connected} \iff \mathrm{M}' \text{ is White-connected.}$$

This reproves the sparse paving case via **circuit-hyperplane relaxations**.

Roadmap

White's conjecture

Stressed subset relaxation

Main theorems and paving matroids

Recent developments

Conclusion

Matroid basis polytopes

From bases to a polytope

For $B \subseteq E$, let

$$\mathbf{e}_B := \sum_{i \in B} \mathbf{e}_i \in \{0,1\}^E.$$

The **basis polytope** of a matroid M is

$$P(\mathrm{M}) := \mathrm{conv}\big\{\mathbf{e}_B : B \in \mathcal{B}(\mathrm{M})\big\} \subseteq \mathbb{R}^E.$$

The ambient hypersimplex

Every rank-r matroid basis polytope on E lies inside the basis polytope of the uniform matroid:

$$P(\mathrm{M}) \subseteq \Delta(r, E) := \mathrm{conv}\{\mathbf{e}_B : B \in \tbinom{E}{r}\}.$$

Stressed subsets and the type map

Stressed subset [FS24]

For $S \subseteq E$, set

$$r' := \mathrm{rank}_{\mathsf{M}}(S), \qquad \tau := r - r'.$$

We call S **stressed** if

$$\mathsf{M}|S = \mathsf{U}_{r',S}, \qquad \mathsf{M}/S = \mathsf{U}_{\tau,E\setminus S}.$$

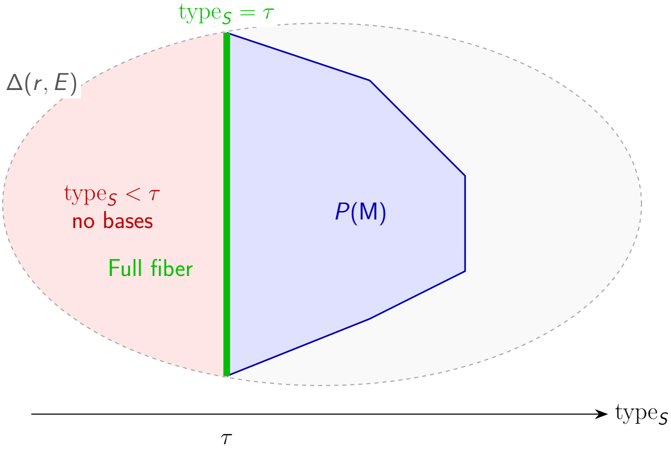
Type map and characterization

Define

$$\mathrm{type}_S(A) := |A \setminus S|.$$

Then S is stressed if and only if:

- **τ-Bounded:** $\mathrm{type}_S(B) \geq \tau$ for every $B \in \mathcal{B}(\mathsf{M})$.
- **Full fiber at τ:** $\mathrm{type}_S^{-1}(\tau) \cap \binom{E}{r} \subseteq \mathcal{B}(\mathsf{M})$.


$\mathrm{type}_S = \tau$
$\Delta(r, E)$
$\mathrm{type}_S < \tau$
no bases
Full fiber
$P(\mathsf{M})$
type_S
τ

Stressed subset relaxation and its specializations

Stressed subset relaxation

A stressed subset provides uniform structures on both S and $E \setminus S$, allowing an explicit enlargement:

$$\mathcal{B}\big(\mathrm{Rel}_S(\mathrm{M})\big) = \mathcal{B}(\mathrm{M}) \cup \left\{ B \in \binom{E}{r} : \mathrm{type}_S(B) < \tau \right\}.$$

Thus, relaxation fills the missing type layers below τ.

A spectrum of relaxations

Object	**New bases**	**Related class**
Stressed subset	Several type layers	Elementary split
Stressed hyperplane	One type layer	**Paving**
Circuit-hyperplane	One new basis	**Sparse paving**

These classes admit sequences of the corresponding relaxations:

$$\mathrm{M} \rightsquigarrow \mathrm{M}_1 \rightsquigarrow \cdots \rightsquigarrow \mathrm{U}_{r,E}.$$

Roadmap

White-connectedness via stressed subsets

Our stressed-subset framework

Building on the circuit-hyperplane relaxation/tightening approach of Han–Michałek–Weigert [HMW25], we obtain:

Relaxation	:	$M \longrightarrow \mathrm{Rel}_S(M)$	preserves White-connectedness,
Tightening	:	$\mathrm{Rel}_S(M) \longrightarrow M$	reduces to degree-2 and degree-3 rerouting.

Application (joint work with Chi Ho Yuen)

The non-commutative version of White's conjecture holds for paving matroids [YY25].

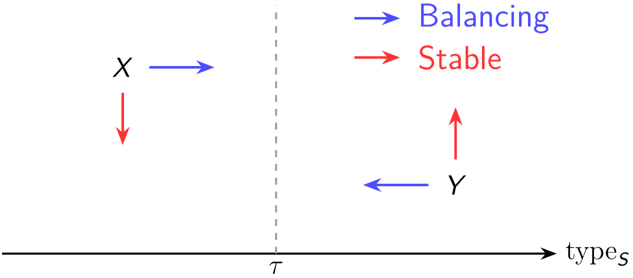
- Paving matroids are conjectured to comprise almost all matroids [Oxl06], and already account for 71.71% of the simple matroids on 9 elements [MR08].
- Partial Steiner systems provide a rich source of paving matroids [HPP22].
- Non-sparse paving matroids are still logarithmically abundant. [Ali26]

Balancing and stable exchanges

Type balancing and stable exchanges [Y.]

For an exchange $(X, Y) \xrightarrow{(x,y)} (X' = X - x + y,\ Y' = Y - y + x)$:

condition	exchange type	effect
$\mathrm{type}(X) < \tau \leq \mathrm{type}(Y)$	$\forall x \exists y;\ \forall y \exists x$	**balance types toward** τ
$\mathrm{type}(X) \leq \tau \leq \mathrm{type}(Y)$	$\forall x \exists y;\ \forall y \exists x$	**type signature unchanged**

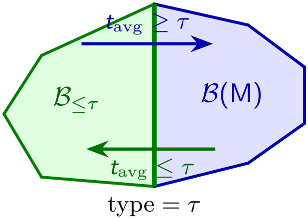


Sketch of SSR preservation theorem

Theorem: SSR preservation [Y.]

$$\mathsf{M} \text{ White-connected} \quad \Longrightarrow \quad \mathrm{Rel}_S(\mathsf{M}) =: \widetilde{\mathsf{M}} \text{ White-connected.}$$

$$\mathbf{B} \approx_{\mathrm{comp}} \mathbf{B}' \Longrightarrow t_{\mathrm{avg}}(\mathbf{B}) := \frac{1}{m}\sum_i \mathrm{type}(B_i) = t_{\mathrm{avg}}(\mathbf{B}')$$



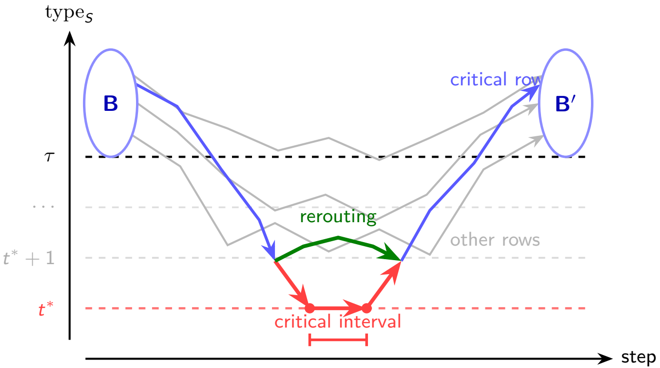
$t_{\mathrm{avg}} \geq \tau$: balance the tuple into $\mathcal{B}(M)$,

$t_{\mathrm{avg}} \leq \tau$: balance the tuple into $\mathcal{B}_{\leq\tau}$.

Sketch of conditional SST preservation

Theorem: Conditional SST preservation [Y.]

If $\widetilde{\mathrm{M}}$ satisfies White's conjecture and the low-degree rerouting postulates hold, then so does M.



The only remaining obstruction is low-degree

Low-degree rerouting postulates

Degree 2 postulate

$$\begin{array}{ccccc} X & \xleftrightarrow{a \quad f} & {\color{red}X''} & \xleftrightarrow{g \quad b} & X' \\ Y & \xleftrightarrow{f \quad a} & Y'' & \xleftrightarrow{b \quad g} & Y' \end{array}$$

$$\mathrm{type}(X'') = t^*;$$

$$\mathrm{type}(X'') + \mathrm{type}(Y'') > 2\tau.$$

Degree 3 postulate

$$\begin{array}{ccccc} X & \xleftrightarrow{a \quad f} & {\color{red}X''} & \xleftrightarrow{g \quad b} & X' \\ Y & \xleftrightarrow{f \quad a} & Y' & & Y \\ Z & & Z & \xleftrightarrow{b \quad g} & Z' \end{array}$$

$$\mathrm{type}(X'') = t^* = \tau - 1$$

$$\mathrm{type}(Y), \mathrm{type}(Z') \geq \tau.$$

There exists a monotone rerouting such that every basis in the X-row has type $> t^*$.

Corollary [Y.]

If White's conjecture holds for $\widetilde{\mathrm{M}}$, the degree-2 and degree-3 cases hold for M, then White's conjecture holds for M.

Application: paving matroids

Theorem [YY25]

The symmetric-exchange version of White's conjecture holds for paving matroids.

Paving matroids:

- Sequence of $\tau = 1$ relaxations: $\mathrm{M} \rightsquigarrow^* \mathrm{U}_{r,E}$.
- Minor-closed (restriction/contraction).
- Every hyperplane (rank $r-1$ flat) is stressed.
- Distinct hyperplanes satisfy $|H \cap H'| \leq r-2$.

Proof idea

- Use minor-closure to remove common and external elements.
- Eliminate degenerate cases; any remaining obstruction forces pairs of hyperplanes to cover E.
- Double-count hyperplane intersections: the covering condition makes the count too large, contradicting $|H \cap H'| \leq r-2$.

Roadmap

Toric ideal viewpoint

Algebraic dictionary

Let $\mathsf{M} = (E, \mathcal{B})$. The **matroid toric ideal** is

$$\mathbf{I}_{\mathsf{M}} = \ker\left(\mathbb{K}[y_B : B \in \mathcal{B}] \to \mathbb{K}[x_e : e \in E]\right), \qquad y_B \mapsto x^B := \prod_{e \in B} x_e.$$

- $\mathbf{B}, \mathbf{B}'$ are compatible $\iff y^{\mathbf{B}} - y^{\mathbf{B}'} \in \mathbf{I}_{\mathsf{M}}$
- Symmetric exchange $\rightsquigarrow$ $y_A y_B - y_{A'} y_{B'} \in \mathbf{I}_{\mathsf{M}}$.

Symmetric exchange White's conjecture

The ideal $\mathbf{I}_{\mathsf{M}}$ is generated by **symmetric exchange quadrics**.

Quadratic White's conjecture: equivalent formulation

The matroid toric ideal $\mathbf{I}_{\mathsf{M}}$ is generated by quadratic binomials

$$y_A y_B - y_{A'} y_{B'} \in \mathbf{I}_{\mathsf{M}}, \qquad A \uplus B = A' \uplus B'.$$

Variants of the conjecture and recent developments

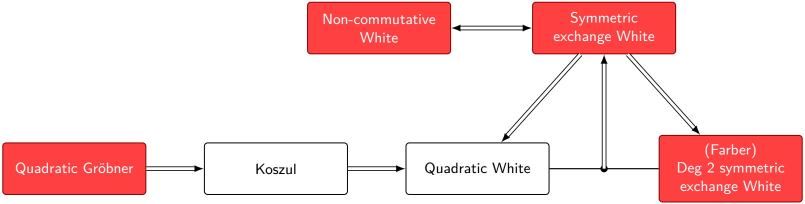
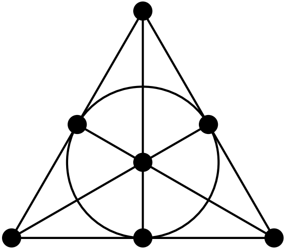


Figure: Implications between variants of White's conjecture. Statements known to be false are colored red. Adapted from Larson [Lar26].

- **Quadratic Gröbner basis: false.**
  The Fano matroid F_7 is a counterexample [Loe+26; Bac+26](9 June 2026; 11 June 2026).
- **Degree-two symmetric-exchange version: false.**
  Larson constructed a rank-9 binary counterexample on 18 elements [Lar26] (2 July 2026).

The Fano matroid F_7



- Rank-3 sparse paving matroid arising from the Steiner system $S(2,3,7)$, with 28 non-collinear 3-subsets as bases.
- Hayase–Hibi–Katsuno–Shibata (2020): all matroids on at most 7 elements were settled, except F_7 and F_7^*.
- De Loera et al. and Backman et al. (2026): both use SAT/SMT search to show that the Fano matroid admits no quadratic Gröbner basis.

Larson's counterexample

$$
\left(\begin{array}{cccccc|cccccc|cccccc}
0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\
\hline
1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1
\end{array}\right)
$$

This rank-9 binary matroid on 18 elements has a disconnected degree-two symmetric-exchange graph.

Larson's ChatGPT 5.5 Pro prompt (abridged)

Find a connected simple matroid M whose graph of partitions

$$E(M) = B_1 \sqcup B_2$$

into two bases, with edges given by symmetric exchanges, is disconnected.

Hints: No example is elementary split (hence none is paving), and none has rank $r \leq 6$. Look for a rank-9 binary matroid represented by

$$[\, I_9 \mid A \,], \qquad A \in \mathrm{GL}_9(\mathbb{F}_2).$$

Your paper certainly helped with the search: knowing where not to look was very useful. Also, I think that perhaps the best chance of finding a counterexample to the quadratic generation version is exhaustive search. There is a new database of (5, 10) matroids (https://matroids.icarm.cloud/). Checking quadratic generation for all 3.2 trillion matroids directly is of course impossible (checking all (9,4) matroids took several hours on my laptop), but it may be feasible if we eliminate cases where the conjecture is known (especially paving matroids).

Figure: Larson's message explaining how our paving result helped narrow the counterexample search.

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After Larson's counterexample

Although the symmetric-exchange version fails in general, our paving result suggests that White-connectedness may still be widespread. It is natural to ask which matroid classes satisfy it.

Our perspective

- We expect the symmetric-exchange version of White's conjecture to hold for split matroids, but finer structural insight into their basis exchanges seems necessary; see [Bér+23].
- The SST argument suggests adding new bases one type layer at a time, potentially extending the stressed-subset relaxation framework to broader classes of split matroids.

Thank you